MATH 261, Homework Set 7 — Answers

1. $\nabla \cdot \mathbf{F} = 4y$; curl k-component $= 0$; conservative: yes.
2. $\nabla \cdot \mathbf{F} = 2x\cos(x^2) + 2ye^{y^2}$; curl k-component $= 1$; conservative: no.
3. $\nabla \cdot \mathbf{F} = 2x$; $\nabla \times \mathbf{F} = (x - 2z, -y, 0)$; conservative: no.
4. $\nabla \cdot \mathbf{F} = 0$; $\nabla \times \mathbf{F} = (0, -3x^2, 0)$; conservative: no.
5. $\nabla \cdot \mathbf{F} = 0$; $\nabla \times \mathbf{F} = (0, 0, 0)$; conservative: yes.
6. $\nabla \cdot \mathbf{F} = -xz\cos y$; $\nabla \times \mathbf{F} = (0, 0, 0)$; conservative: yes.
7. At $(1, -1, 1)$: curl $= (2, 0, 0)$; divergence $= 7$.
8. At $(1, -1, 1)$: curl $= (0, 3, 4)$; divergence $= -9$.
9. $f = x^2 + 3y + 2z^2 + C$.
10. $f = xy^2 + ye^z + C$.
11. $f = x\cos z + y^2 + C$.
12. 16.
13. 0.
14. Direct line integral: 2π; Green's theorem: 2π.
15. Direct line integral: 0; Green's theorem: 0.
16. $\dfrac{1}{2}$.
17. $\dfrac{7}{6}$.
18. $3(26 - e^5)$.
19. $\dfrac{9}{2}(e^5 + 9)$.
20. -1.
21. $-\dfrac{1}{3}$.
22. $-\dfrac{5}{12}$.
23. $-\dfrac{1}{3}$.
24. 0.
25. $\dfrac{22}{15}$.

Worked solutions: Problems 7, 21, and 23

7. Curl and divergence at $(1,-1,1)$.

Write $\mathbf{F} = (P,Q,R) = (x^2z, -2y^2z^2, xy^2z)$. Then

$$\nabla\cdot\mathbf{F} = P_x + Q_y + R_z = 2xz - 4yz^2 + xy^2,$$
$$(\nabla\cdot\mathbf{F})(1,-1,1) = 2+4+1 = \boxed{7}.$$

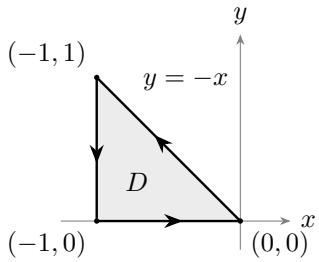
The curl is

$$\begin{aligned}\nabla\times\mathbf{F} &= (R_y - Q_z,\ P_z - R_x,\ Q_x - P_y)\\ &= (2xyz - (-4y^2z),\ x^2 - y^2z,\ 0-0),\end{aligned}$$
$$(\nabla\times\mathbf{F})(1,-1,1) = (-2+4,\ 1-1,\ 0) = \boxed{(2,0,0)}.$$

21. Outward flux across the triangle from Problem 20.

Here $\mathbf{F} = (P,Q) = (x^2+y, y^2-x)$. The shaded triangle D has bounds

$$\begin{cases} -1 \le x \le 0, \\ 0 \le y \le -x. \end{cases}$$



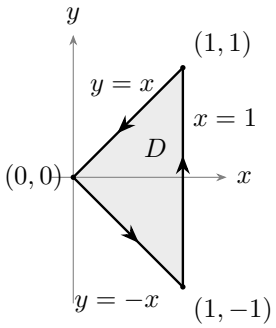
For the outward unit normal $\mathbf{N}$, Green's flux formula uses $P_x + Q_y = 2x + 2y$. Thus

$$\oint_C \mathbf{F}\cdot\mathbf{N}\,ds = \iint_D (P_x+Q_y)\,dA = \int_{-1}^{0}\int_{0}^{-x} (2x+2y)\,dy\,dx$$
$$= \int_{-1}^{0} [2xy + y^2]_0^{-x}\,dx = \int_{-1}^{0} -x^2\,dx = \left[-\frac{x^3}{3}\right]_{-1}^{0} = \boxed{-\frac{1}{3}}.$$

23. Work around the triangle between $y=-x$ and $y=x$.

The traversal $(0,0)\to(1,-1)\to(1,1)\to(0,0)$ is counterclockwise. The shaded region D has bounds

$$\begin{cases} 0 \le x \le 1, \\ -x \le y \le x. \end{cases}$$



For $P = x^2y$ and $Q = xy^2$, we have $Q_x - P_y = y^2 - x^2$. Using the circulation form of Green's theorem,

$$W = \oint_C P\,dx + Q\,dy = \int_0^1\int_{-x}^{x} (y^2-x^2)\,dy\,dx$$
$$= \int_0^1 \left[\frac{y^3}{3} - x^2y\right]_{-x}^{x} dx = \int_0^1 -\frac{4}{3}x^3\,dx = \left[-\frac{x^4}{3}\right]_0^1 = \boxed{-\frac{1}{3}}.$$